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# Shaping the Journey: How GenAI-enabled Mobile Interfaces Transform Visitor Experiences in Tourism Contexts

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**Abstract**
Drawing on Gestalt theory and cognitive load theory, this paper takes an exploratory qualitative approach to understanding users' perceptions and expectations of GenAI-enabled mobile apps for tourism. Methodologically, 27 users of travel apps were interviewed using semi-structured interviews to evaluate visual and functional aspects of a conceptual app design. Results showed that users have service quality expectations, visual appeal, and enjoyment, driven by intuitive navigation, visual consistency, and cognitive load. Visual design was found to be an aesthetic attribute, with expected enjoyment related to simplicity, interactivity, and clarity. Findings suggest that users assess such technologies and form expectations based on the anticipated interplay among visual, functional, and interactive elements. This paper bridges perceptual and cognitive approaches to inform tourism research by showing how conceptual app designs can evoke user expectations early in innovation processes. Results offer a theoretically grounded guide to GenAI-enabled tourism app designs that align with user expectations, enabling experience transformation in emerging tourism markets.

**Keywords** Generative artificial intelligence · Digital tourism innovation · Visitor-centered design · Suez Canal region · Customized experiences

## 1 Introduction

The rapid development of mobile apps has transformed destination planning and experiences to support tourism (Chon & Hao, 2025). It has provided traveling visitors with real-time information and personalized directions like never before (Selem & Selim, 2025; Sia et al., 2024). This change marks a new phase of intelligent tourism, driven by the use of artificial intelligence (AI) and natural language processing (NLP) tools. Therefore, this provides the necessary capabilities for creating interactive and personalized experiences (Chon & Hao, 2025). Given that AI is integrated into tourism destination management and tourism service delivery, it reforms visitors' interactions with tourism destinations (Gajić et al., 2024).

Despite technological improvements, gaps remain in the existing tourism literature (Saghier et al., 2026). Most studies on GenAI-enhanced travel apps have been conducted in major destinations that attract large numbers of foreign tourists (Florido-Benítez & del Alcazar Martinez, 2024; Topsakal & Çuhadar, 2025) and have focused on informational or transactional apps. Nevertheless, the cognitive and experiential effects on visitors when content is delivered through interactive interfaces are less considered in such apps. Few empirical studies have examined how visitors consider such designs in new destinations (Gajić et al., 2024; Selem & Selim, 2025). This leaves a knowledge gap regarding the skills of generative AI (GenAI) for use as a communication and interaction tool in underrepresented communities (Ilieva et al., 2024). The Suez Canal is not digitized in Egypt, and tourism is not developed through digital platforms (El-Masri, 2024), while other iconic destinations (e.g., Cairo, Luxor, and Sharm El-Sheik) are prioritized. This paper bridges the gap by discussing visitor feedback on a GenAI-enabled mobile app design for this little-known area.

Our theoretical framework comprises Gestalt theory (GST) and cognitive load theory (CLT). GST focuses on the visual structure and holistic perception and proposes that interface participants should interpret its elements

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as unified patterns (Wang et al., 2024). The proposed app design uses GST to apply to illustrations, icons, and colors to boost first impressions and easy navigation. Conversely, CLT highlights the use of lower values of cerebral energy for digital content processing (Martínez-Moleles et al., 2024). Therefore, CLT-inspired designs make decision-making easier and require less psychological effort. This paper blends visual appeal and cognitive effectiveness by integrating GST and CLT. Thus, this ensures that GenAI-generated illustrations are engaging and can inform decision-making. This paper makes three contributions to the tourism technology literature. First, it contributes to understanding how GenAI-enabled mobile interfaces affect user experiences during the pre-implementation phase, bridging a gap in existing research that has focused on post-adoption behaviors. Second, it combines GST and CLT to understand the combined effects of perceptual and cognitive processes on user expectations in digital tourism. Third, it offers an empirically-informed conceptualization of GenAI-enabled mobile app design for underrepresented tourism destinations.

With such theoretical assumptions, this paper has two objectives. First, this paper focuses on how visitors perceive GenAI-enabled visual and interactive elements in mobile tourism apps. Second, GenAI-enabled app design is tested to foster sustainable tourism perceptions amongst underserved tourism destinations. Port Said, at the entrance to the Mediterranean, is a city rich in culture and marine heritage. Ismailia is also a promising destination for cultural and heritage tourism, thanks to its archaeological sites and festivals. In addition, Suez is a city with attractions associated with its seaside beauty and its industrial and military history. Such cities represent untapped potential in Egyptian tourism. Therefore, they are also an exciting study population for investigating innovative GenAI-enabled solutions to enhance visibility and cultural interactions in this region.

## 2 Literature Review

### 2.1 Underpinned Theories

GST offers a strong foundation for examining visitors' perception and use of GenAI-enabled mobile interfaces in tourism contexts. GST suggests that visitors do perceive visual features in isolation and integrated wholes, guided by proximity, similarity, and continuity in their perception and interpretation (Wang et al., 2024). In tourism contexts, such principles are embodied in interface elements (e.g., icons, layout, and color), affecting initial impressions, ease of use, and engagement (Bilgihan et al., 2024; Ilieva et al., 2024). Specifically, well-designed interfaces help users understand complex tourism-related information, enabling decision-making (Joseph et al., 2024). GST proposes that visually cohesive and culturally relevant designs can alter tourism content perceptions (Selem & Selim, 2025), leading to increased trust, satisfaction, and engagement (Chen & Chen, 2025).

CLT also complements GST by providing valuable insights into how users process information in digital spaces and the need to reduce cognitive load. Tourism apps provide users with extensive content on destinations, attractions, and services (Saghier et al., 2026), which may overwhelm them if poorly organized (El-Akhras et al., 2025). CLT breaks down cognitive load into intrinsic, extraneous, and germane, suggesting that user interfaces should minimize extraneous load by presenting content and structuring information. Earlier studies have emphasized the importance of intuitive navigation, visual design elements, and organized layouts in improving cognitive efficiency and user satisfaction in mobile apps (Huang et al., 2024; Srinivasan et al., 2024). Regarding GenAI-enabled tourism apps, CLT suggests that personalized suggestions, interactive elements, and efficient navigation can help users' process information, enhancing engagement and satisfaction (Selem & Selim, 2025). GST and CLT offer a holistic view of how visual design and cognitive processing interact to affect visitor experiences in mobile apps.

### 2.2 GenAI in Tourism Applications

Tourism is experiencing rapid change as GenAI is adopted to create personalized, scalable, and interactive digital experiences (Saghier et al., 2026). GenAI-enabled apps (e.g., chatbots and virtual assistants support trip planning and interactions with services) enhance visitor satisfaction (Florido-Benítez & del Alcázar Martínez, 2024). Tourism websites (e.g., Booking.com and Expedia) incorporate AI-enabled features to provide personalized experiences, exemplifying the growing trend toward smart, data-driven tourism systems (Gajić et al., 2024). Over-reliance on AI-enabled tools can diminish human contact, a key component of hospitality operations (Vashishth et al., 2024). Likewise, data bias, fake content, and privacy concerns can erode user trust in AI-generated content (El-Akhras et al., 2025). Earlier research has emphasized the need to balance technological advancements with ethical and user-focused approaches to integrate GenAI in tourism sustainably (Gajić et al., 2024; Khan & Khan, 2024). Although GenAI has the potential to transform tourism experiences, its success depends on seamless integration into cognitively efficient interfaces.

## 2.3 Visitor Perceptions in Digital Tourism Environments

Tourist perceptions are crucial in the success of online tourism apps (Saghier et al., 2026). Previous studies have highlighted factors that play a crucial role in shaping user experience (e.g., service quality, visual appeal, and perceived enjoyment) and how such factors contribute to user engagement and behavioral intentions (Huang et al., 2024; Sia et al., 2024). Earlier research has highlighted the pivotal roles of such factors in users' perceptions and use of mobile tourism apps.

### 2.3.1 Service Quality Expectations

Users' expectations of service quality in digital tourism apps relate to the speed, accuracy, and effectiveness of the apps in meeting travel-related needs (Huang et al., 2024). In mobile tourism, service quality is associated with the provision of reliable information, ease of navigation, and the ability to access multiple services through a single interface, which in turn leads to higher user satisfaction and trust (Bilgihan et al., 2024). Existing research has highlighted that intuitive interfaces and real-time services (e.g., booking, location-based services, and personalized recommendations) are essential for supporting decision-making and enhancing user satisfaction (Khan & Khan, 2024). Tourism planning can be complex and information-heavy; users demand that apps ease this process through accessible and trustworthy services. From this perspective, service quality perceptions focus on functional aspects and how information is presented to users. As such, service quality is accepted as a critical predictor of user satisfaction, trust, and loyalty towards digital tourism services (Huang et al., 2024). In GenAI-enabled mobile interfaces, such expectations are heightened (Saghier et al., 2026). Customers expect integrated travel planning and tourism experiences.

### 2.3.2 Perceived Enjoyment

Perceived enjoyment describes how users feel appreciative and satisfied with using mobile apps, beyond their functional advantages (Huang et al., 2024). In tourism contexts, enjoyment is an essential attribute of user interaction, as it increases users' desire to navigate content, use features, and continue using mobile apps (Sia et al., 2024). Prior research has shown that interactive features, customized suggestions, and aesthetically pleasing user interfaces can improve users' emotional reactions and overall experience (Saghier et al., 2026; Vashishth et al., 2024). Specifically, crafted digital spaces that combine interactivity and simplicity can promote positive user experiences and reduce cognitive load. As a result, perceived enjoyment is one of the main factors predicting future use of apps and continued use of tourism apps (Sia et al., 2024). Therefore, enjoyment is likely to be a byproduct of immersive content, intuitive navigation, and user-focused design in GenAI-enabled mobile interfaces (Saghier et al., 2026), affecting how visitors use and perceive tourism websites.

### 2.3.3 Visual Appeal

Visual appeal refers to the aesthetic and structural features of digital interfaces (e.g., layout design, color scheme, typography, and graphical elements) that affect how users perceive and interact with apps (Chen & Chen, 2025). Visual appeal is also essential for creating initial impressions, perceived usability, and credibility in tourism apps, as users tend to judge an app's quality by its visual appearance before they can use its features (Srinivasan et al., 2024). Topsakal & Çuhadar (2025) also highlighted the importance of visually consistent, clean interfaces for improving user interaction by helping users navigate with ease and reducing cognitive load. In addition, attractive visuals can trigger emotions, arouse curiosity, and prompt users to delve deeper into tourism content (Joseph et al., 2024). Visual appeal does operate solely at the aesthetic level and supports cognitive processing by organizing information in easily accessible ways. Consequently, the effectiveness of user experience and decision-making can be enhanced with well-designed visual interfaces (Saghier et al., 2026). Considering GenAI in mobile apps, visual appeal is achieved through dynamic, AI-generated content that can enhance user experiences and be rich without sacrificing consistency and clarity.

## 2.4 Conceptualization of the GenAI-enabled App Design

Expanding on the integration of GST and CLT, the proposed GenAI-enabled mobile app adopts a visitor-centered design strategy to create and enhance tourism experiences through structured, user-friendly, and visually consistent interfaces. The proposed app, Explore Suez Canal Cities, is designed as a GenAI-enabled mobile platform that combines functionality with dynamic content creation to aid exploration, planning, and interaction in underrepresented tourism destinations. Interfaces consist of basic elements, including destination browsing, classified tourism activities, real-time events and festivals, interactive maps, and reservation systems, arranged in a single, accessible structure. Earlier research has

highlighted the importance of integrating visual design and functionality to enhance usability, engagement, and satisfaction on online tourism platforms (Ilieva et al., 2024; Srinivasan et al., 2024). In this regard, the proposed app design takes a holistic approach, integrating visual, functional, and interactive features to deliver a smooth user experience.

Theoretically, GST explains the visual structuring of interfaces to achieve coherence, consistency, and perceptual clarity via structured layouts, recognizable icons, and balanced color schemes. Simultaneously, CLT helps simplify the interaction streams of information, reducing cognitive load and promoting effective decision-making. By incorporating such theoretical concepts, the proposed design shapes how users perceive, process, and interact with tourism information, affecting their overall experience. This is a conceptual app used as an exploratory model to explain how GenAI-enabled interfaces can transform visitors' experiences by aligning design features with expectations. Figure 1 presents our conceptual framework and shows how GenAI-enabled mobile interfaces affect visitor experiences through interface design grounded in GST and CLT. This shows that such processes mediate the main dimensions of customer experiences, shaping visitor experiences. Figure 2 extends the conceptual framework by showing the proposed GenAI-enabled tourism apps, including their structure and functionality. It shows how the frontend user interfaces, back-end AI components, and data integration layers work to provide personalized, interactive tourism experiences.

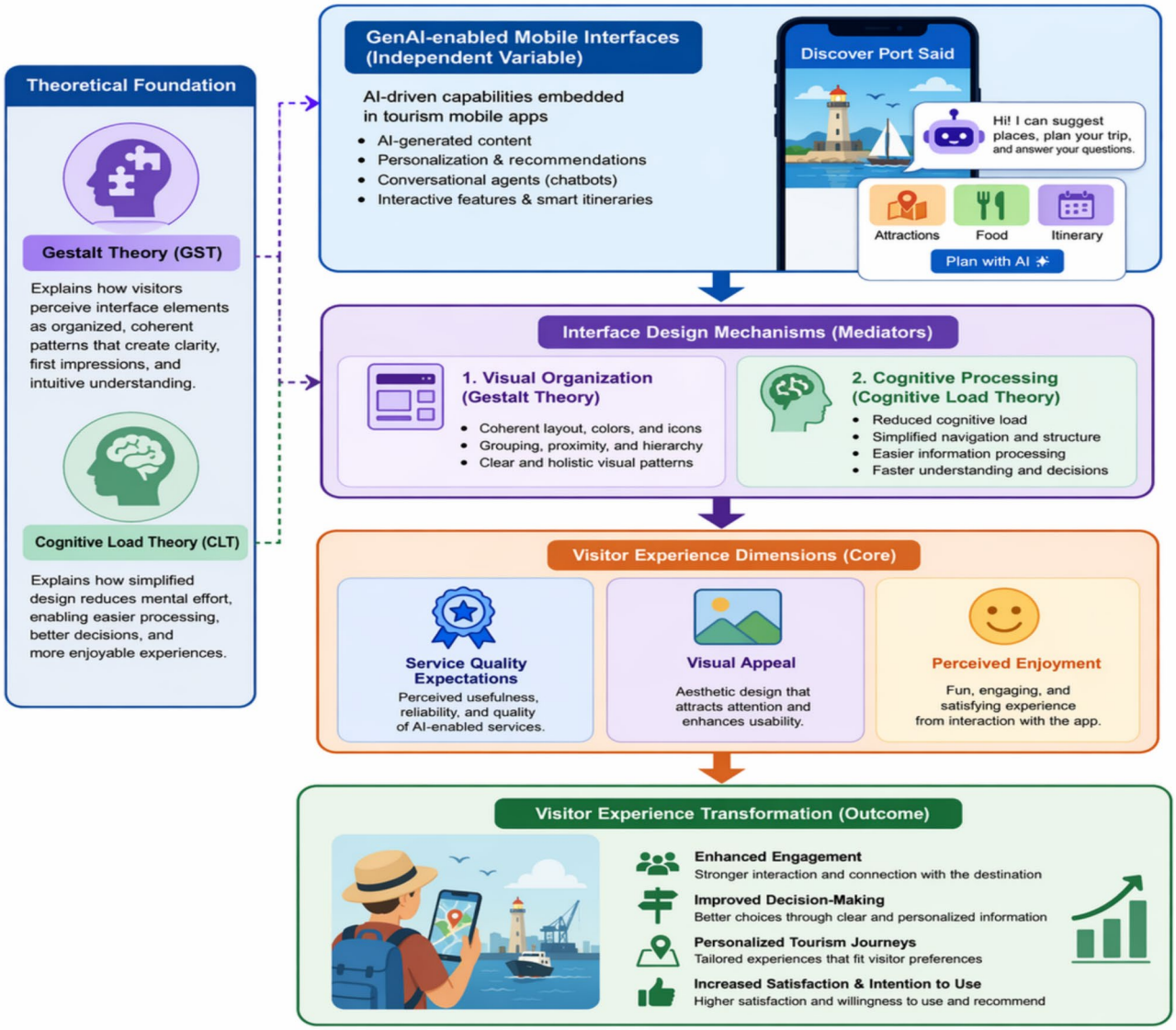



**Fig. 1** Conceptual framework of how GenAI-enabled mobile interfaces shape and transform visitor experiences.**Source:** Author preparation

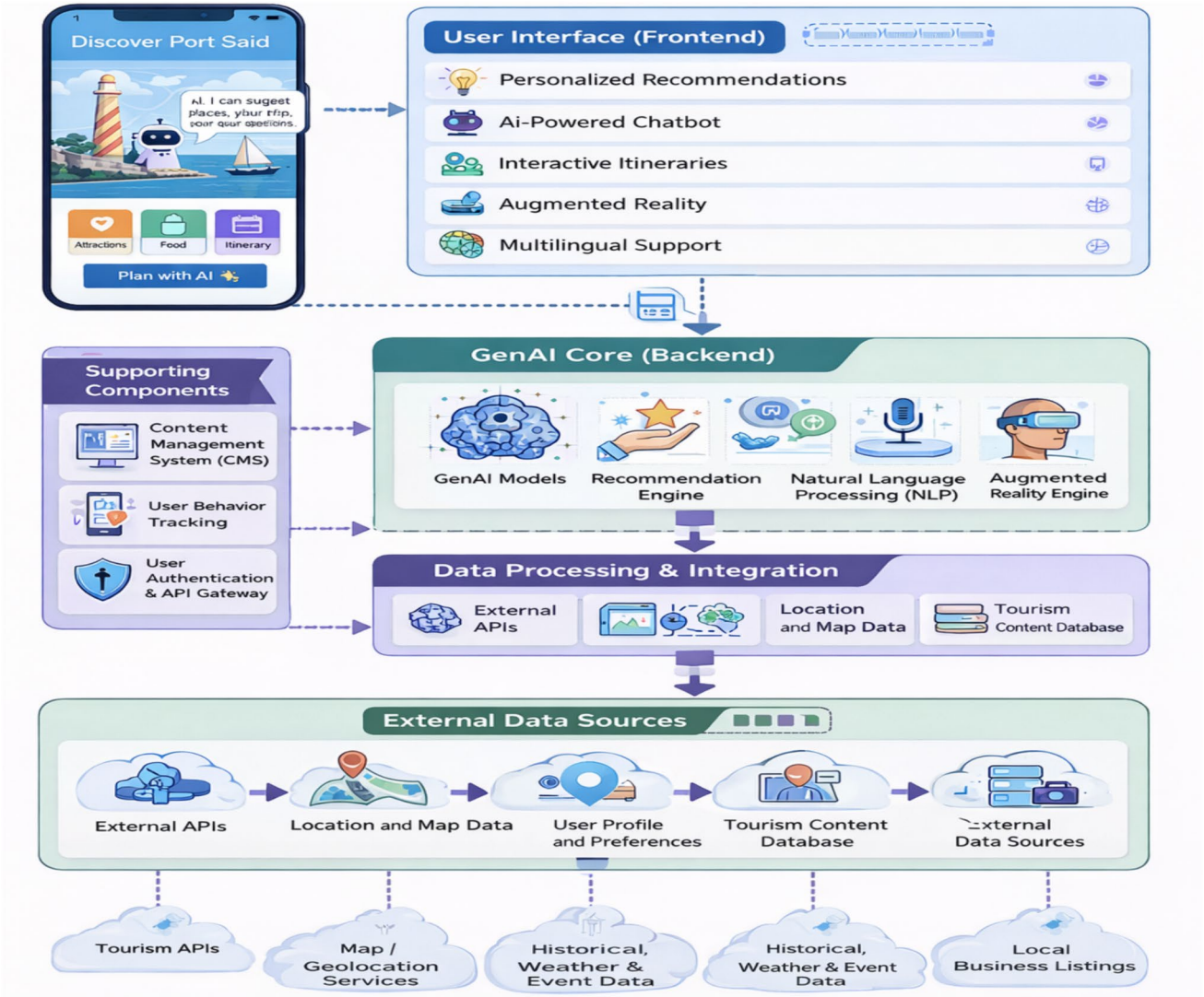



**Fig. 2** Structure and functional architecture of the proposed GenAI-enabled tourism application.**Source:** Author preparation

## 3 Methods

### 3.1 Research Design and Approach

This paper adopts an exploratory qualitative design to investigate how GenAI-enabled mobile interfaces may shape and transform visitor experiences in tourism contexts. Since the proposed app is conceptual and this paper is dedicated to future perceptions and expectations, a qualitative method is suitable for gaining rich, context-specific insights that would be unattainable with quantitative approaches. A post-positivist epistemology underlies this paper, presupposing the systematic exploration of user perceptions through structured data collection and comprehensive analysis. Semi-structured interviews were chosen because they offered the opportunity to reflect on the proposed app design.

### 3.2 Sampling Strategy and Participant Recruitment

A purposive sampling approach was employed, with participants selected for their ability to provide informed, relevant information about mobile tourism apps. This method is recommended in qualitative studies, where the goal is to acquire information-rich cases aligned with this paper's objectives. The eligibility criteria included that participants must be aged 18 years, fluent in English or Arabic, and have experience using travel-enabled mobile apps. Such conditions ensured that participants were digitally literate and familiar with assessing conceptual app designs. Participants were located through professional networks and social media, as well as through those with experience in the tourism, academic, and business fields. Invitations were sent via email and direct messages, with participation requirements

and the promise of confidentiality and voluntary participation. A total of 27 individuals accepted to participate (response rate: 68% of the 40 visitors invited). This sample size is in line with the requirements of qualitative research, which emphasizes depth of insight.

### 3.3 Data Collection Procedures and Ethical Considerations

Data were collected through semi-structured online Zoom interviews in December 2024 and February 2025. Interviews lasted about 25–40 min each, which was enough for an in-depth discussion. Before interviews, participants were introduced to a conceptual description and visual mock-ups of the proposed GenAI-enabled tourism app (see Appendix 1). To ensure consistency across sessions, interviewers provided a standard explanation of apps' features. Three main constructs included in the interview guide (service quality expectations, visual appeal, and perceived enjoyment) were organized around open-ended questions that were intended to help the individual reflect on such constructs (see Appendix 2). Whenever needed, follow-up probes were used to clarify answers and gain deeper insights. All interviews were audio-recorded with interviewees' consent, and field notes were made to provide contextual observations and improve data interpretation. This paper was conducted voluntarily, and participants were aware of this paper's purpose, confidentiality, and their freedom to withdraw at any time. No sensitive personal information was collected, and during transcription and analysis, all responses were anonymized. Formal ethical approval was not required under the institution's guidelines for non-invasive research.

### 3.4 Data Analysis

Audio tapes were transcribed with Leximancer v.5, and qualitative data were handled and analyzed with QSR NVivo (v.11). The analysis followed a rigorous six-step thematic approach. First, data familiarization was conducted by repeatedly reading transcripts and notes. Second, the first codes were created based on deductive constructs grounded in GST and CLT, as well as inductive knowledge emerging from participant feedback. Third, the codes were categorized into nodes and were organized into larger themes that represented recurring patterns. Fourth, themes were narrowed down and tested to provide conceptual clarity and internal consistency. Fifth, the definitions of each theme were developed and linked to the conceptual framework. Lastly, results were analyzed against the theoretical frameworks outlined in this paper. Thematic constructs were defined to ensure that data collection and analysis were on the same wavelength.

### 3.5 Data Saturation, Trustworthiness, and Rigor

Data collection continued until thematic saturation was achieved, defined as the point at which no new meaningful insights or patterns emerged from additional interviews. Participants' responses were also consistent in the late phases of data collection, indicating that datasets were rich and deep. The sample size of 27 participants is also in line with previous qualitative studies in tourism research, which also achieved saturation to a similar extent. Various strategies were used to guarantee the credibility, confirmability, and transferability of our findings. Credibility was bolstered by the long-term work with datasets, repetitive coding, and the use of verbatim quotations from participants to support interpretations. Reliability was ensured through a systematic analysis process, including clear records of coding procedures and the development of themes. Confirmability was also addressed by grounding findings in participant data and by reducing researcher bias through reflexive interpretation.

## 4 Findings

Figure 3 presents the thematic framework of visitor expectations for GenAI-enabled mobile interfaces, highlighting the interdependence of service quality expectations, visual appeal, and perceived enjoyment as the key dimensions that determine the anticipated user experience. The interviews lasted 25 to 40 min, during which participants were allowed to share detailed thoughts on their experience with travel apps and how they viewed the proposed app design. The sample was composed of women (70.4%) and males (29.6). The age structure shows digitally savvy demographics, with most participants aged 25–35 (48.2%), 18–25 (22.2%), 35–45 (18.5%), and 45 + (11.1%). The sample was diverse in terms of nationalities, with Egyptians (37%), Indians (18.5%), Greeks (11.1%), Saudis (7.4%), Italians (7.4%), and Omani (7.4%). It provided a range of cultural attitudes relevant to tourism consumption in the Suez Canal region. The sample consisted of professionals in tourism (44.4%), academics (37%), business professionals (14.8%), and other professions (3.7%), offering practical and theoretical insights into app design and usability. Participants were assessing GenAI-enabled mobile app interfaces. Thus, results indicate perceived expectations, usefulness expectations, and design preferences.

### 4.1 Theme 1: Service Quality Expectations

Participants reiterated the importance of the app's intuitive, efficient, and structured design. Under GST, participants




**Fig. 3** Thematic structure of visitor expectations toward GenAI-enabled mobile interfaces.**Source:** Author preparation

shared the view that a clear grouping of interface elements, similar iconography, and a logical navigation structure would make it easier to recognize and interact with mobile app interfaces. Simultaneously, CLT-inspired expectations highlighted the need to minimize cognitive load by simplifying layouts, presenting information, and providing smooth access to essential functions, including activity schedules and recommendations. Some participants have stated that the proposed design meets their overall expectations for tourism apps in terms of integrating multiple services into a single interface. Fourteen participants noted that a well-designed graphical interface is essential to appeal to international visitors. Besides, seven participants stressed that it is important to be able to investigate tourism activities and cities without difficulty, indicating expectations for easy navigation and a well-organized content architecture.

Another point raised by participants was the presence of well-built reservation functions, which would facilitate proper booking processes, indicating that it was necessary to be clear and comprehensive in transaction features. Hence, attributes were critical in perceived service quality. Sixteen participants also suggested that hotels, restaurants, and tourism services would be more accessible with labeled icons and page navigation tabs. Nonetheless, minor usability issues were noted regarding font size and readability. Participants viewed the proposed app as becoming holistic and rich in information, with 21 participants indicating that such designs would offer comprehensive, customized tourism information. This was demonstrated by one of the participants, who said, "It would be better than before since the app could help organize my travel plans and make everything easy to locate" (Participant 12). Such observations illustrate how participants relate effective service quality to organized design, cognitive simplicity, and integrated functionality.

### 4.2 Theme 2: Visual Appeal

Visual appeal was found to be a decisive factor in how participants expect to be engaged with the proposed app. According to GST, participants highlighted the importance of visual cohesion (e.g., consistent color, familiar icons, and symmetrical designs) to ensure instant comprehension and favorable initial impressions. Participants suggested that clear interfaces and visually guided navigation would diminish cognitive load and increase interpretability. Eleven participants were optimistic about using vivid colors and culturally appropriate illustrations, which they believed would add aesthetic value and attraction to tourism destinations. Meanwhile, 16 participants cited areas for improvement, the need for consistent design patterns across cities, and a lack of visual differentiation. Nine participants also highlighted the importance of a well-designed home interface in minimizing psychological effort by conveying the app's functionality. This is indicative of visual hierarchy in helping cognitive processing and user orientation. Participant 7 observed: I would like to have more distinct city visuals so that each destination is unique in design systems. Nineteen participants said the proposed illustrations should be modern and eye-catching, and sixteen said visual elements would help shape the app's identity.

### 4.3 Theme 3: Perceived Enjoyment

Perceived enjoyment is the expected emotional and experience reactions of participants to the suggested app design. In line with GST, participants stated that well-integrated, aesthetically pleasing interfaces would foster a sense of completeness and simplicity in interaction. At the same time, expectations based on CLT placed particular emphasis on avoiding extraneous cognitive load to keep users engaged and prevent user fatigue. Participants proposed that AI-generated drawings, interactive travel plans, and customized suggestions could increase enjoyment by providing natural experiences. Seventeen participants anticipated the app would be enjoyment and user-friendly, and twenty-five emphasized the importance of a well-defined, easy-to-understand visual introduction to facilitate initial interaction. With such positive expectations, there were concerns about the potential complexity of visuals. Three participants found that a conflicting design may add to cognitive load and decrease clarity. This highlights the need to strike a balance between visual richness and simplicity. Another point highlighted by participants was that a structured presentation of content and a hierarchical information design would allow them to focus on pertinent information, improving their enjoyment.

The expected purpose of enjoyment in maintaining interest is evident in participant responses, like: "In case the app is easy, interactive, and visual, I would like to use it and spend more time on it" (Participant 19). Moreover, twenty-six participants proposed that the visual and interactive elements would be well-designed and thus interesting and encouraging, and seven participants mentioned that such features would be a value-added to their tourism exploration. Overall, visual clarity, interactivity, and ease of use were correlated with perceived enjoyment, which supports the supporting nature of GST and CLT in positive user expectations. Such themes showed that visitor expectations are not constructed in a vacuum and are functional, visual, and experiential interaction results. Visitor perceptions of service quality, visual attractiveness, and perceived enjoyment contribute to how users expect to interact with the proposed app, underscoring the need to integrate usability, aesthetics, and interactivity into GenAI-enabled tourism interfaces.

## 5 Discussion and Conclusions

### 5.1 General Discussion

This paper offers a new understanding of the effects of GenAI-enabled mobile interfaces on visitor experiences in tourism settings by analyzing users' anticipatory assessments of hypothetical application plans. Results should be viewed as an indicator of expected views, since participants rated an imaginary design. Within this exploration framing, findings indicate that interface design elements based on GST and CLT enhance service quality expectations, visual appeal, and enjoyment, and provide clues on how GenAI-enabled apps can change tourism experiences during the design phase. Figure 4 demonstrates how GST and the CLT can be integrated to affect visitor experiences. It shows the mutual effects of perceptual organization and cognitive processing on user interaction, decision-making, and experience transformation in GenAI-enabled tourism interfaces.

First, results showed that GenAI-enabled mobile interfaces are likely to revolutionize visitor experiences, as they affect how users engage with information and services in tourism. Participants were also keen on aspects (e.g., smooth navigation, built-in features, and organized presentation of the content). Such expectations imply that users want interfaces that make decision-making easier and simplify the process, which aligns with CLT principles that focus on minimizing cognitive load (Martínez-Mololes et al., 2024). Participants explained how such features would affect their experience by rendering travel planning more effective, convenient, and consistent. GST regarded organized layouts

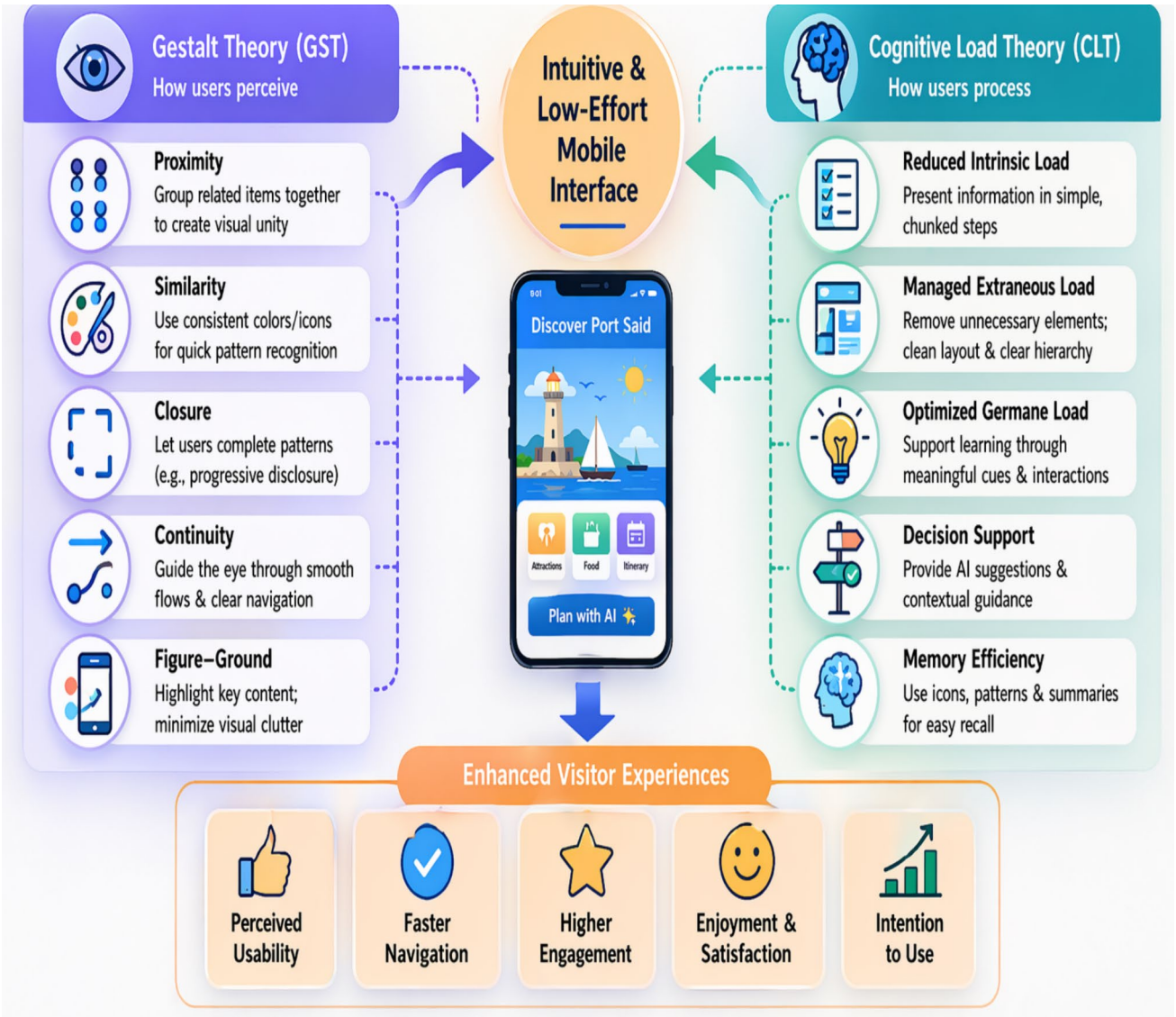



**Fig. 4** Integration of Gestalt theory and cognitive load theory in shaping visitor experiences.**Source:** Author preparation

and regular interface elements as mechanisms to affect user perception and interaction by organizing visual groupings and recognizing patterns (Wang et al., 2024). Therefore, GenAI-enabled design is likely to transform disorganized data into structured, valuable tourism experiences.

Second, findings demonstrate that visual design is a key factor in shaping and altering visitor experiences, extending beyond aesthetics to include usability and interaction. Coherent visual structures, clear iconography, and consistent color schemes were synonymous with participants, with ease of readability and reliability. Such results support previous studies showing that visual attractiveness affects engagement and perceived value in tourism apps (Srinivasan et al., 2024). Visually organized interfaces shape perception by allowing users to organize intricate information into cohesive patterns (Chen & Chen, 2025). At the same time, CLT describes how organized visual contexts reduce cognitive burden, influencing smoother, more efficient user experiences (Martínez-Mololes et al., 2024). Participants demanded that the use of further visual distinction implies that the transformation of visitor experiences needs coherence, but also uniqueness associated with the identity of a certain destination.

Third, results indicate that GenAI-enabled mobile interfaces are likely to reshape visitor experiences by providing increased enjoyment, arising from the balance between interactivity and cognitive simplicity. Participants suggested that more enjoyable tourism experiences could be achieved through AI-generated content, personalized recommendations, and interactive itineraries. However, this is hedonic and determined by the degree to which interfaces are understandable and cognitively accessible. This is consistent with

previous literature, which has attributed perceived enjoyment to engagement and retention (Huang et al., 2024; Sia et al., 2024). Under CLT, enjoyment is determined by the optimal load of cognition, and the GST can clarify how designed visual interfaces can increase affective responses through perceptual fluency (Chen & Chen, 2025). The worries expressed by participants regarding visual complexity suggest that a poorly balanced design can be an obstacle to experience transformation by placing higher cognitive burdens.

Moreover, results showed that GenAI-enabled mobile interfaces can change visitors' experience by shaping their perceptions, processing, and engagement with information. This paper demonstrates that visitors can form holistic expectations about how they anticipate visual, functional, and interactive elements to interact. GST and CLT integration offers a powerful explanatory framework for how such transformations occur, with particular focus on the interaction between perceptual organization and cognitive processing. By doing so, this paper adds to the research on technology in tourism by demonstrating that experience transformation can be initiated at expectation levels. Lastly, this paper indicates that conceptual app designs can shape initial visitor expectations and educate them about how emerging technologies could alter tourism experiences in new destinations. This supports the need for AI-enabled systems that incorporate transparency, consistency, and interaction to meet user expectations. Future studies ought to build on such lessons by investigating how the envisaged changes are translated into practical behaviors and experiences as a single operational system is put into practice.

## 5.2 Theoretical Implications

This paper contributes to tourism and hospitality research by providing a conceptually grounded insight into how visitors develop expectations for GenAI-enabled mobile apps during the pre-implementation phase. This paper addresses existing demands to explore cognitive and interpretive facets of GenAI-enabled tourism apps through nuanced research (Gajić et al., 2024; Ilieva et al., 2024) by grounding findings in perceptions. This paper shows that visitors develop service quality evaluations, visual appeal, and enjoyment despite interacting with conceptual app images. This is applicable in emerging tourism contexts, where functional systems may be lacking. However, decisions related to strategic design should be mindful of user expectations. One of the key theoretical contributions of this paper is the integration of GST and CLT to understand how users cognitively engage with AI-enabled interfaces and how they evaluate them. Although previous studies of tourism have focused on such theories separately (Martínez-Mololes et al., 2024; Wang et al., 2024), this paper demonstrates their complementary roles in shaping user expectations.

GST describes how visitors process visual information by perceiving visual element patterns as coherent, affecting perceived clarity and attractiveness. Simultaneously, CLT provides cognitive grounding, encouraging users to consider the mental effort required to process information, which is necessary for reducing extraneous load through an organized, simplified design. Integrating such views, this paper presents a comprehensive cognitive-perceptual framework that reveals that anticipated usability and engagement result from the interplay between visual organization and cognitive efficiency. In addition, this paper adds to the literature by re-conceptualizing visual appeal as a functional and cognitive theory. The available literature on tourism has tended to consider visual appeal as a subordinate determinant of attractiveness or emotional reaction (Joseph et al., 2024; Saghier et al., 2026). However, our results show that participants view the visual design as part of the navigation and decision-making processes. GST can improve perceptual fluency, whereas CLT indicates that well-organized visual objects can reduce cognitive load and facilitate the efficient processing of information (Chen & Chen, 2025). This expands existing frameworks by making visual appeal a key mechanism linking interface design to cognitive performance and user expectations in GenAI-enabled tourism apps.

Moreover, this paper contributes to theoretical knowledge of perceived enjoyment in digital tourism settings by demonstrating that it relies on cognitive simplicity and organized interactivity. Although previous literature has already determined the role of enjoyment in technology adoption (Huang et al., 2024; Sia et al., 2024), this paper narrows this association by revealing that hedonic or entertainment features trigger enjoyment and participants in question equated enjoyment with interfaces that were interactive and understandable, as well as minimized mental load. At the same time, GST describes how affective responses are promoted by perceptual completeness in visually harmonious and integrated interfaces (Wang et al., 2024). Thus, this paper hypothesizes that perceived enjoyment in GenAI-enabled apps is cognitively mediated, resulting in engagement, simplicity, and visual coherence.

Lastly, this paper provides a theoretical and methodological expansion to early-phase innovation studies in tourism technology. This paper shows the importance of perception-based assessments in underdeveloped areas, demonstrating that conceptual app designs can evoke meaningful, structured user expectations. This aligns with the broader tourism innovation literature, which highlights the significance of design-oriented methods for learning about emerging technologies (Florido-Benítez & del Alcázar Martínez,

2024; Gajić et al., 2024). Our results indicate that pre-use cognitive and perceptual evaluations should be considered a separate phase in theoretical models of technology adoption and user experience. Therefore, this paper helps to fill the gap between the conceptual design and empirical validation in underrepresented tourism destinations where digital transformation is still developing (El-Masri, 2024).

### 5.3 Practical Implications

Our results have practical implications for tourism stakeholders by revealing that GenAI-enabled mobile interface design can shape visitor expectations and interactions with digital tourism platforms. User-centric app adoption, incorporating smooth navigation, visually appealing layouts, and interactive features, should be a priority for tourism authorities and destination management organizations. Such apps can make tourism content interfaces user-friendly by aligning interface design with visitors' expectations for service quality, visual appeal, and enjoyment. GenAI-enabled systems offer personalized, context-sensitive information, enabling visitors to explore tourism destinations and activities based on their preferences. This makes decision-making processes easier and enhances overall satisfaction and perceived value of tourism experiences in underrepresented destinations. Governments and regulatory authorities are critical policy stakeholders in ensuring the development of GenAI-enabled tourism apps. Tourism providers can be motivated to pursue digital transformation efforts through financial incentives and supportive regulatory frameworks.

Accessibility, including multilingualism and culturally sensitive information, should also be a focus of regulatory guidelines to ensure inclusivity for various visitor groups. Such measures help tourism authorities ensure that GenAI apps enhance visitor experiences and meet international accessibility, transparency, and ethical design standards. Results also indicated the significance of cross-sector cooperation in the effective implementation of GenAI-enabled tourism solutions. Therefore, digital platforms need to be coordinated with app developers, tourism businesses, cultural heritage managers, and regional government units. This can help introduce platforms that offer integrated services (e.g., accommodation or transportation). Additionally, GenAI-enabled apps can help manage visitor flows and enhance destination efficiency. Tourism authorities can make sound decisions to allocate resources, develop infrastructure, and plan destinations by leveraging data-driven insights generated by such apps, encouraging sustainable tourism development.

Lastly, GenAI-enabled mobile apps have significant consequences for capacity building and workforce development in tourism contexts. Stakeholders and governments must consider investing in training programs to promote digital literacy with the knowledge to use AI-enabled tools. Such efforts can facilitate the adoption of innovative technologies across the hotel, restaurant, and transportation sectors, enabling businesses to be responsive to changes in visitor expectations. Moreover, performance benchmarks and accreditation programs can be implemented to assess and encourage the usage of high-quality digital solutions. Tourism stakeholders can ensure that GenAI-enabled apps contribute to long-term improvements in visitor satisfaction and destination competitiveness by developing a technologically competent workforce.

### 5.4 Limitations and Future Research

Although this paper makes several contributions, it has several limitations that offer future research directions. First, the empirical situation is confined to Suez Canal, which might limit the applicability of our results to other tourism destinations with different cultural, technological, and infrastructural factors. Future research should expand this paper to other geographic settings to examine the viability of proposed design principles across diverse tourism contexts. Second, this paper is founded on a GenAI-enabled mobile app. Further studies are needed to build and test functioning GenAI-enabled tourism apps to confirm these results using an experimental or longitudinal design. Future research must include interdisciplinary collaboration with software developers and designers to develop working systems that enable realistic user testing. Such developments would enable researchers to analyze user behavior, usage habits, and decision-making in real time. Lastly, GenAI-enabled mechanisms for content validation and multilingual assistance might improve the accuracy, reliability, and cultural sensitivity of tourism apps, enhancing their effectiveness for diverse global users.

## Appendix 1 Description of the Conceptual GenAI-enabled Mobile Application

Participants were presented with a conceptual GenAI-enabled mobile tourism application titled Explore Suez Canal Cities, designed to enhance visitor experiences in underrepresented tourism destinations. This app was not fully functional; instead, it was introduced through a combination of visual mock-ups and a structured verbal explanation provided during the interview. The application was described as an AI-driven platform that integrates generative AI to deliver personalized, interactive, and visually engaging tourism experiences. Its core purpose is to

assist users in exploring destinations, planning itineraries, and accessing tourism-related services through a unified digital interface. The app interface included several key components:

- **Home page**: A central dashboard presenting featured destinations, highlighted attractions, special offers, and upcoming events. The interface adapts to user preferences through AI-generated recommendations.
- **Tourism cities section**: A dedicated space showcasing major cities within the Suez Canal region (e.g., Port Said, Ismailia, and Suez), each represented through AI-generated illustrations and curated content.
- **Tourism activities**: A categorized section allowing users to explore activities based on themes such as cultural, historical, natural, and shopping experiences.
- **Events and festivals**: A dynamic interface displaying real-time updates on local events, schedules, and ticket availability.
- **Interactive map**: A navigation tool designed to help users locate attractions, accommodations, and services.
- **Reservation function**: A feature intended to allow users to book hotels, activities, and services through the app, with detailed information on availability and pricing.
- Exploration and Itineraries: AI-generated travel suggestions, including personalized day-by-day itineraries based on user preferences.
- **Visual design**: The app incorporates a consistent visual identity using color themes representing different tourism elements (e.g., blue for coastal tourism, green for cultural landscapes, and yellow for desert environments). Icons, layouts, and illustrations were designed to be intuitive and visually cohesive.
- **Virtual assistant**: An AI-powered chatbot providing support, answering queries, and guiding users through the platform. Participants were asked to evaluate the app based on these descriptions and visual representations. As such, their responses reflect anticipated expectations, perceived usefulness, and design preferences.

**Source**: Author preparation.

## Appendix 2 Semi-structured Interview Guide

This paper employed a semi-structured interview approach to explore participants' perceptions of a conceptual GenAI-enabled mobile tourism application. The interview guide was designed based on the study's conceptual framework and focused on three core dimensions: service quality expectations, visual appeal, and perceived enjoyment. Open-ended questions were used to encourage rich, reflective responses, while follow-up probes were applied where necessary to deepen understanding.

### Section A: Background and Prior Experience

- How do you use mobile applications when planning or managing your travel?
- What types of travel apps have you used before, and what features do you find most useful?
- Can you describe a positive or negative experience you have had with tourism mobile apps?

### Section B: Perceptions of the Proposed App Design

Participants were first introduced to the conceptual app design through visual mock-ups and verbal explanation:

- Based on the presented app design, how do you expect this application to support your travel planning and tourism activities?
- What aspects of the app design do you find most useful or valuable?
- Are there any features that you feel are missing or could be improved?
- Can you elaborate on why this feature is important?
- How would this influence your travel decisions?

### Section C: Service Quality Expectations

- How do you perceive the app's ability to provide reliable and accurate tourism information?
- To what extent do you expect the app to be easy to navigate and use?
- How important is it for you that the app integrates multiple services (e.g., booking, maps, activities) in one place?
- What are your expectations regarding responsiveness and real-time updates (e.g., events, weather, availability)?
- What would make the app more efficient for you?
- How should information be organized to improve your experience?

### Section D: Visual Appeal

- How do you evaluate the visual design of the app, including colors, icons, and layout?
- Do you find the interface clear and visually appealing? Why or why not?

- How important is visual design in influencing your decision to use a tourism app?
- Are there any visual elements you would change?
- Do the visuals help you understand the app easily?

### Section E: Perceived Enjoyment

- To what extent do you think using this app would be enjoyable or engaging?
- What features would make the app more interactive or interesting for you?
- How likely are you to explore tourism destinations using such an app?
- What would make the experience more fun or engaging?
- Would you spend more time using this app? Why?

### Section F: Overall Evaluation and Future Expectations

- Overall, how would you describe your impression of this app design?
- What improvements would you recommend to enhance user experience?
- Would you consider using such an app if it were developed? Why or why not?

**Source**: Author preparation.

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## Declarations

**Conflict of interest** The author declares no competing interests.

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